## 11.1 Parametrizations of Plane Curves

### Parametric Equations

Figure 11.1 shows the path of a moving particle in the $xy$-plane. Notice that the path fails the vertical line test, so it cannot be described as the graph of a function of the variable $x$. However, we can sometimes describe the path by a pair of equations, $x = f(t)$ and $y = g(t)$, where $f$ and $g$ are continuous functions. When studying motion, $t$ usually denotes time. Equations like these can describe more general curves than those described by a single function, and they provide not only the graph of the path traced out but also the location of the particle $(x, y) = (f(t), g(t))$ at any time $t$.

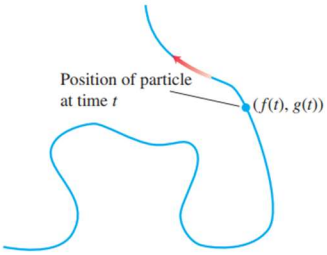



**FIGURE 11.1** The curve or path traced by a particle moving in the $xy$-plane is not always the graph of a function or single equation.

**DEFINITION** If $x$ and $y$ are given as functions

$$x = f(t), \qquad y = g(t)$$

over an interval $I$ of $t$-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a **parametric curve**. The equations are **parametric equations** for the curve.

**EXAMPLE 1** Sketch and identify the curve defined by the parametric equations

$$x = t^2 - 2t \qquad y = t + 1$$

**EXAMPLE 2** What curve is represented by the following parametric equations?

$$x = \cos t \qquad y = \sin t \qquad 0 \leq t \leq 2\pi$$

## EXAMPLE 1 An Entire Parabola

The position $P(x, y)$ of a particle moving in the $xy$-plane is given by the equations and parameter interval

$$x = t, \qquad y = t^2, \qquad -\infty < t < \infty.$$

Identify the particle's path and describe the motion.

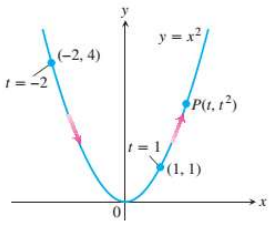



**FIGURE 10.28** The path defined by $x = t, y = t^2, -\infty < t < \infty$ is the entire parabola $y = x^2$ (Example 1).

**EXAMPLE 2** Sketch the curve defined by the parametric equations

$$x = t^2, \qquad y = t + 1, \qquad -\infty < t < \infty.$$

**EXAMPLE 5** A parametrization of the graph of the function $f(x) = x^2$ is given by

$$x = t, \qquad y = f(t) = t^2, \qquad -\infty < t < \infty.$$

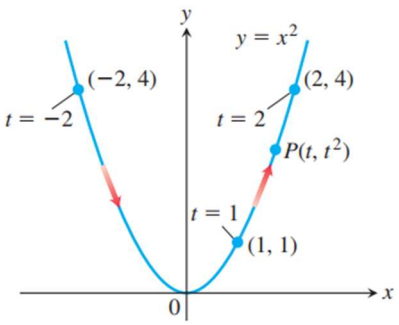



**FIGURE 11.6** The path defined by $x = t, y = t^2, -\infty < t < \infty$ is the entire parabola $y = x^2$ (Example 5).

## 11.2 Calculus with Parametric Curves

In this section we apply calculus to parametric curves. Specifically, we find slopes, lengths, and areas associated with parametrized curves.

### Tangents and Areas

A parametrized curve $x = f(t)$ and $y = g(t)$ is **differentiable** at $t$ if $f$ and $g$ are differentiable at $t$. At a point on a differentiable parametrized curve where $y$ is also a differentiable function of $x$, the derivatives $dy/dt$, $dx/dt$, and $dy/dx$ are related by the Chain Rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

If $dx/dt \neq 0$, we may divide both sides of this equation by $dx/dt$ to solve for $dy/dx$.

**Parametric Formula for $dy/dx$**
If all three derivatives exist and $dx/dt \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}. \tag{1}$$

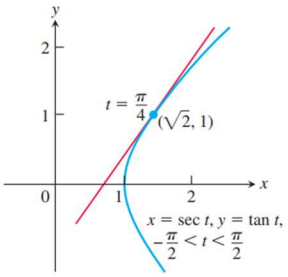



**FIGURE 11.14** The curve in Example 1 is the right-hand branch of the hyperbola $x^2 - y^2 = 1$.

**Parametric Formula for $d^2y/dx^2$**
If the equations $x = f(t)$, $y = g(t)$ define $y$ as a twice-differentiable function of $x$, then at any point where $dx/dt \neq 0$ and $y' = dy/dx$,

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}. \tag{2}$$

**EXAMPLE 1** Find the tangent to the curve

$$x = \sec t, \qquad y = \tan t, \qquad -\frac{\pi}{2} < t < \frac{\pi}{2},$$

at the point $\left(\sqrt{2}, 1\right)$, where $t = \pi/4$ (Figure 11.14).

**EXAMPLE 2** Find $d^2y/dx^2$ as a function of $t$ if $x = t - t^2$ and $y = t - t^3$.

**EXAMPLE 3** Find the area enclosed by the astroid (Figure 11.15)

$$x = \cos^3 t, \qquad y = \sin^3 t, \qquad 0 \le t \le 2\pi.$$

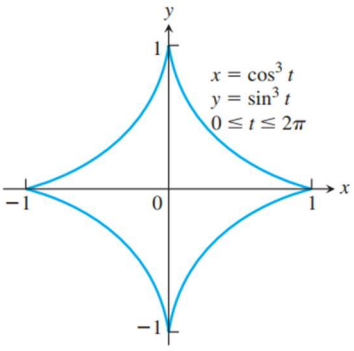



**FIGURE 11.15** The astroid in Example 3.

## Length of a Parametrically Defined Curve

Let $C$ be a curve given parametrically by the equations

$$x = f(t) \qquad \text{and} \qquad y = g(t), \qquad a \le t \le b.$$

**DEFINITION** If a curve $C$ is defined parametrically by $x = f(t)$ and $y = g(t)$, $a \le t \le b$, where $f'$ and $g'$ are continuous and not simultaneously zero on $[a, b]$, and $C$ is traversed exactly once as $t$ increases from $t = a$ to $t = b$, then **the length of $C$** is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt.$$

If $x = f(t)$ and $y = g(t)$, then using the Leibniz notation we can write the formula for arc length this way:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt. \tag{3}$$

**EXAMPLE 4** Using the definition, find the length of the circle of radius $r$ defined parametrically by

$$x = r\cos t \qquad \text{and} \qquad y = r\sin t, \qquad 0 \le t \le 2\pi.$$

## Areas of Surfaces of Revolution

In Section 6.4 we found integral formulas for the area of a surface when a curve is revolved about a coordinate axis. Specifically, we found that the surface area is $S = \int 2\pi y\, ds$ for revolution about the $x$-axis, and $S = \int 2\pi x\, ds$ for revolution about the $y$-axis. If the curve is parametrized by the equations $x = f(t)$ and $y = g(t)$, $a \le t \le b$, where $f$ and $g$ are continuously differentiable and $(f')^2 + (g')^2 > 0$ on $[a, b]$, then the arc length differential $ds$ is given by Equation (4). This observation leads to the following formulas for area of surfaces of revolution for smooth parametrized curves.

### Area of Surface of Revolution for Parametrized Curves

If a smooth curve $x = f(t)$, $y = g(t)$, $a \le t \le b$, is traversed exactly once as $t$ increases from $a$ to $b$, then the areas of the surfaces generated by revolving the curve about the coordinate axes are as follows.

**1. Revolution about the $x$-axis ($y \ge 0$):**

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \tag{5}$$

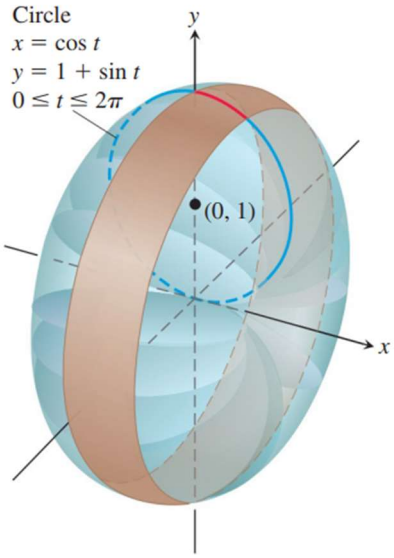
**2. Revolution about the $y$-axis ($x \ge 0$):**

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \tag{6}$$

**EXAMPLE 9** The standard parametrization of the circle of radius 1 centered at the point (0, 1) in the $xy$-plane is

$$x = \cos t, \qquad y = 1 + \sin t, \qquad 0 \le t \le 2\pi.$$

Use this parametrization to find the area of the surface swept out by revolving the circle about the $x$-axis (Figure 11.19).



# 11.3 Polar Coordinates

## Definition of Polar Coordinates

To define polar coordinates, we first fix an **origin** $O$ (called the **pole**) and an **initial ray** from $O$ (Figure 11.20). Usually the positive $x$-axis is chosen as the initial ray. Then each point $P$ can be located by assigning to it a **polar coordinate pair** $(r, \theta)$ in which $r$ gives the directed distance from $O$ to $P$ and $\theta$ gives the directed angle from the initial ray to ray $OP$. So we label the point $P$ as

$P(r, \theta)$

Directed distance from $O$ to $P$ — Directed angle from initial ray to $OP$

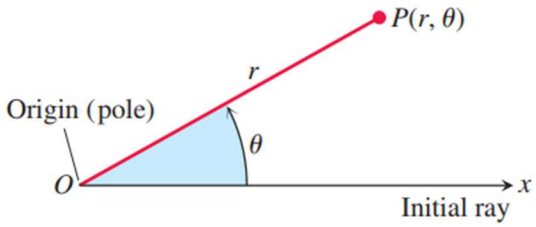



**FIGURE 11.20** To define polar coordinates for the plane, we start with an origin, called the pole, and an initial ray.

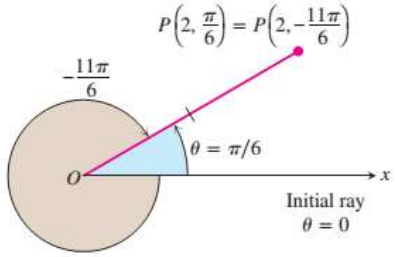



**FIGURE 10.36** Polar coordinates are not unique.

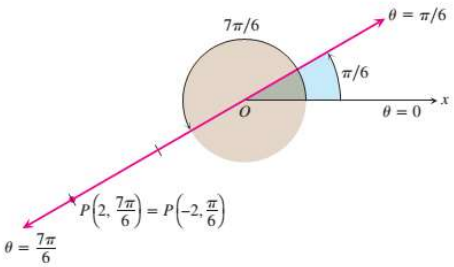



**FIGURE 10.37** Polar coordinates can have negative $r$-values.

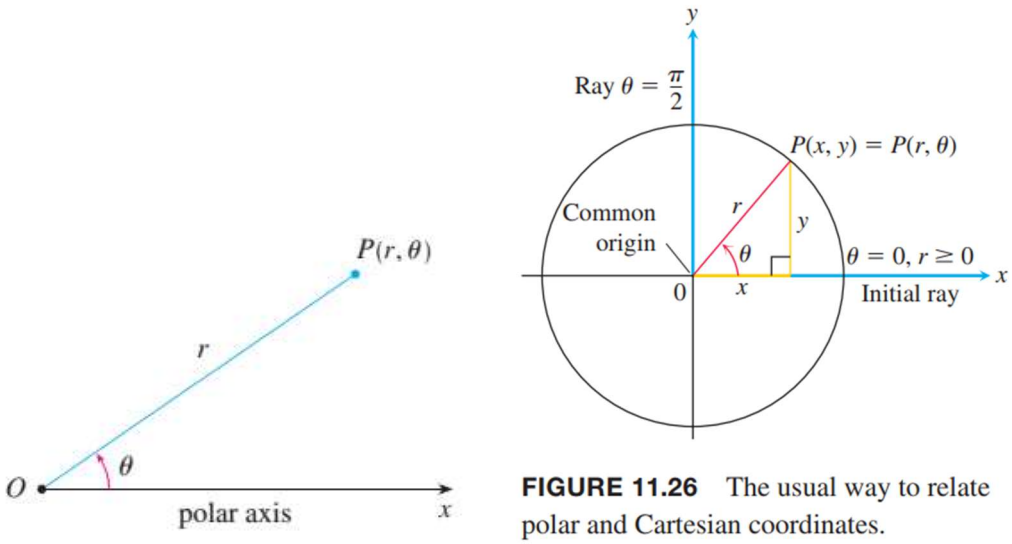



**FIGURE 11.26** The usual way to relate polar and Cartesian coordinates.

**Equations Relating Polar and Cartesian Coordinates**

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r^2 = x^2 + y^2, \qquad \tan\theta = \frac{y}{x}$$

## EXAMPLE 1 Finding Polar Coordinates

Find all the polar coordinates of the point $P(2, \pi/6)$.

**EXAMPLE 4** Here are some plane curves expressed in terms of both polar coordinate and Cartesian coordinate equations.

| Polar equation | Cartesian equivalent |
|---|---|
| $r\cos\theta = 2$ | $x = 2$ |
| $r^2\cos\theta\sin\theta = 4$ | $xy = 4$ |
| $r^2\cos^2\theta - r^2\sin^2\theta = 1$ | $x^2 - y^2 = 1$ |
| $r = 1 + 2r\cos\theta$ | $y^2 - 3x^2 - 4x - 1 = 0$ |
| $r = 1 - \cos\theta$ | $x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$ |

Some curves are more simply expressed with polar coordinates; others are not. ■

**EXAMPLE 5** Find a polar equation for the circle $x^2 + (y - 3)^2 = 9$ (Figure 11.27).

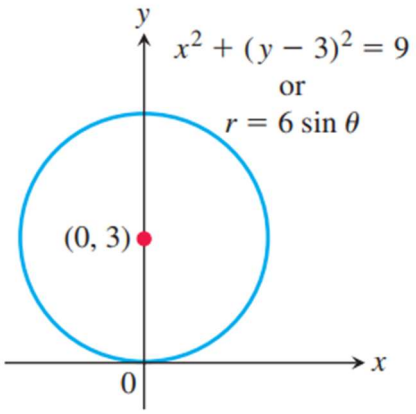



**FIGURE 11.27** The circle in Example 5.

**EXAMPLE 2** Finding Polar Equations for Graphs

**(a)** $r = 1$ and $r = -1$ are equations for the circle of radius 1 centered at $O$.

**(b)** $\theta = \pi/6$, $\theta = 7\pi/6$, and $\theta = -5\pi/6$ are equations for the line in Figure 10.38.

**EXAMPLE 6** Replace the following polar equations by equivalent Cartesian equations and identify their graphs.

**(a)** $r \cos \theta = -4$

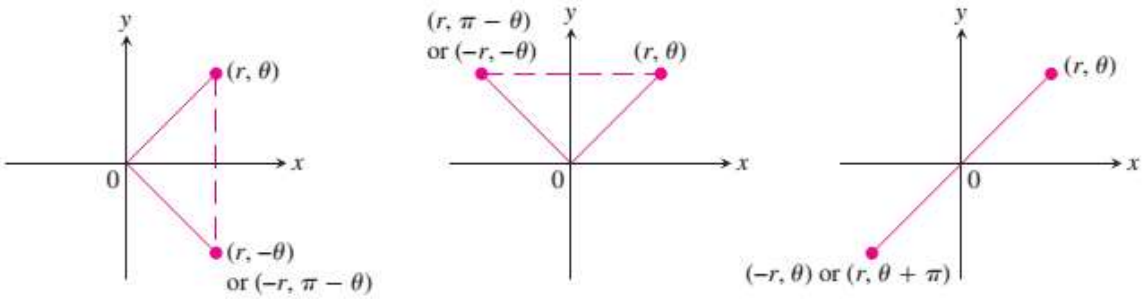
**(b)** $r^2 = 4r \cos \theta$

**(c)** $r = \dfrac{4}{2 \cos \theta - \sin \theta}$

# 11.4 Graphing Polar Coordinate Equations

## Symmetry

Figure 10.43 illustrates the standard polar coordinate tests for symmetry.




(a) About the $x$-axis (b) About the $y$-axis (c) About the origin

### Symmetry Tests for Polar Graphs in the Cartesian $xy$-Plane

1. *Symmetry about the x-axis:* If the point $(r, \theta)$ lies on the graph, then the point $(r, -\theta)$ or $(-r, \pi - \theta)$ lies on the graph (Figure 11.28a).
2. *Symmetry about the y-axis:* If the point $(r, \theta)$ lies on the graph, then the point $(r, \pi - \theta)$ or $(-r, -\theta)$ lies on the graph (Figure 11.28b).
3. *Symmetry about the origin:* If the point $(r, \theta)$ lies on the graph, then the point $(-r, \theta)$ or $(r, \theta + \pi)$ lies on the graph (Figure 11.28c).

## Slope

The slope of a polar curve $r = f(\theta)$ is given by $dy/dx$, not by $r' = df/d\theta$. To see why, think of the graph of $f$ as the graph of the parametric equations

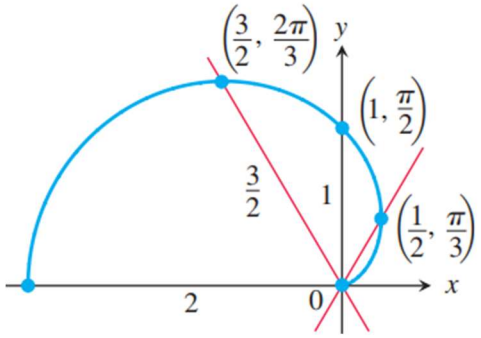
$$x = r\cos\theta = f(\theta)\cos\theta, \qquad y = r\sin\theta = f(\theta)\sin\theta.$$

### Slope of the Curve $r = f(\theta)$ in the Cartesian $xy$-Plane

$$\left.\frac{dy}{dx}\right|_{(r,\,\theta)} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta} \tag{1}$$

provided $dx/d\theta \neq 0$ at $(r, \theta)$.

**EXAMPLE 1** Graph the curve $r = 1 - \cos\theta$ in the Cartesian $xy$-plane.




**EXAMPLE 6**

(a) Sketch the curve with polar equation $r = 2\cos\theta$.

(b) Find a Cartesian equation for this curve.

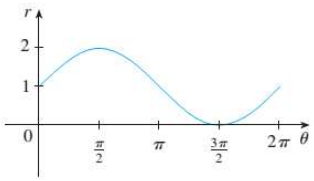



**FIGURE 10**
$r = 1 + \sin\theta$ in Cartesian coordinates, $0 \leqslant \theta \leqslant 2\pi$

**EXAMPLE 7** Sketch the curve $r = 1 + \sin\theta$.

SOLUTION Instead of plotting points as in Example 6, we first sketch the graph of $r = 1 + \sin\theta$ in *Cartesian* coordinates in Figure 10 by shifting the sine curve up one unit. This enables us to read at a glance the values of $r$ that correspond to increasing values of $\theta$. For instance, we see that as $\theta$ increases from 0 to $\pi/2$, $r$ (the distance from $O$) increases from 1 to 2, so we sketch the corresponding part of the polar curve in Figure 11(a). As $\theta$ increases from $\pi/2$ to $\pi$, Figure 10 shows that $r$ decreases from 2 to 1, so we sketch the next part of the curve as in Figure 11(b). As $\theta$ increases from $\pi$ to $3\pi/2$, $r$ decreases from 1 to 0 as shown in part (c). Finally, as $\theta$ increases from $3\pi/2$ to $2\pi$, $r$ increases from 0 to 1 as shown in part (d). If we let $\theta$ increase beyond $2\pi$ or decrease beyond 0, we would simply retrace our path. Putting together the parts of the curve from Figure 11(a)–(d), we sketch the complete curve in part (e). It is called a **cardioid**, because it's shaped like a heart.

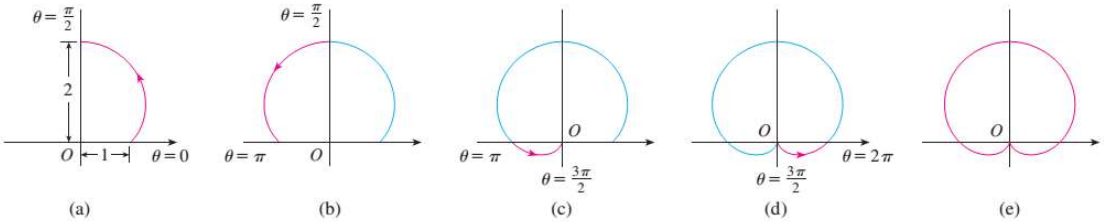



**FIGURE 11** Stages in sketching the cardioid $r = 1 + \sin\theta$

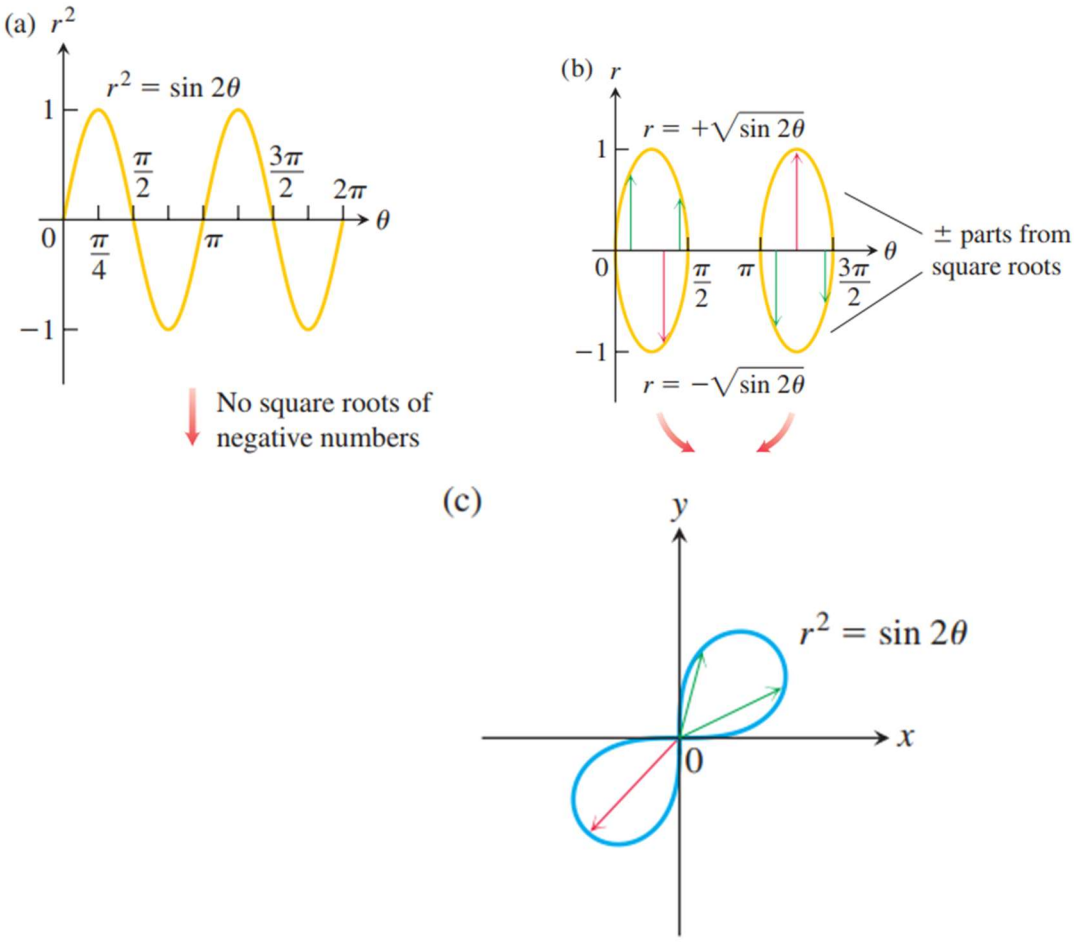
## Converting a Graph from the $r\theta$- to $xy$-Plane

One way to graph a polar equation $r = f(\theta)$ in the $xy$-plane is to make a table of $(r, \theta)$-values, plot the corresponding points there, and connect them in order of increasing $\theta$. This can work well if enough points have been plotted to reveal all the loops and dimples in the graph. Another method of graphing is to

1. first graph the function $r = f(\theta)$ in the *Cartesian* $r\theta$-plane,
2. then use that Cartesian graph as a "table" and guide to sketch the *polar* coordinate graph in the $xy$-plane.

This method is sometimes better than simple point plotting because the first Cartesian graph shows at a glance where $r$ is positive, negative, and nonexistent, as well as where $r$ is increasing and decreasing. Here is an example.

**EXAMPLE 3** Graph the *lemniscate* curve $r^2 = \sin 2\theta$ in the Cartesian $xy$-plane.




**EXAMPLE 9**

(a) For the cardioid $r = 1 + \sin\theta$ of Example 7, find the slope of the tangent line when $\theta = \pi/3$.

(b) Find the points on the cardioid where the tangent line is horizontal or vertical.

## EXERCISES 11.4

### Symmetries and Polar Graphs

Identify the symmetries of the curves in Exercises 1–12. Then sketch the curves in the $xy$-plane.

**1.** $r = 1 + \cos\theta$

**2.** $r = 2 - 2\cos\theta$

**3.** $r = 1 - \sin\theta$

**4.** $r = 1 + \sin\theta$

**5.** $r = 2 + \sin\theta$

**6.** $r = 1 + 2\sin\theta$

**7.** $r = \sin(\theta/2)$

**8.** $r = \cos(\theta/2)$

**9.** $r^2 = \cos\theta$

**10.** $r^2 = \sin\theta$

**11.** $r^2 = -\sin\theta$

**12.** $r^2 = -\cos\theta$

Graph the lemniscates in Exercises 13–16. What symmetries do these curves have?

**13.** $r^2 = 4\cos 2\theta$

**14.** $r^2 = 4\sin 2\theta$

**15.** $r^2 = -\sin 2\theta$

**16.** $r^2 = -\cos 2\theta$

### Slopes of Polar Curves in the $xy$-Plane

Find the slopes of the curves in Exercises 17–20 at the given points. Sketch the curves along with their tangents at these points.

**17. Cardioid** $r = -1 + \cos\theta; \quad \theta = \pm\pi/2$

**18. Cardioid** $r = -1 + \sin\theta; \quad \theta = 0, \pi$

**19. Four-leaved rose** $r = \sin 2\theta; \quad \theta = \pm\pi/4, \pm 3\pi/4$

**20. Four-leaved rose** $r = \cos 2\theta; \quad \theta = 0, \pm\pi/2, \pi$

## 11.5 Areas and Lengths in Polar Coordinates

This section shows how to calculate areas of plane regions and lengths of curves in polar coordinates.

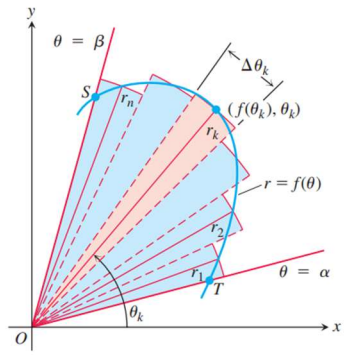



**FIGURE 11.32** To derive a formula for the area of region *OTS*, we approximate the region with fan-shaped circular sectors.

### Area in the Plane

The region *OTS* in Figure 11.32 is bounded by the rays $\theta = \alpha$ and $\theta = \beta$ and the curve $r = f(\theta)$. We approximate the region with $n$ nonoverlapping fan-shaped circular sectors based on a partition $P$ of angle $TOS$. The typical sector has radius $r_k = f(\theta_k)$ and central angle of radian measure $\Delta\theta_k$. Its area is $\Delta\theta_k/2\pi$ times the area of a circle of radius $r_k$, or

$$A_k = \frac{1}{2}r_k^2\,\Delta\theta_k = \frac{1}{2}\big(f(\theta_k)\big)^2\,\Delta\theta_k.$$

The area of region *OTS* is approximately

$$\sum_{k=1}^{n} A_k = \sum_{k=1}^{n} \frac{1}{2}\big(f(\theta_k)\big)^2\,\Delta\theta_k.$$

If $f$ is continuous, we expect the approximations to improve as the norm of the partition $P$ goes to zero, where the norm of $P$ is the largest value of $\Delta\theta_k$. We are therefore led to the following formula for the region's area:

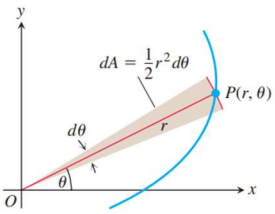



**FIGURE 11.33** The area differential $dA$ for the curve $r = f(\theta)$.

$$A = \lim_{\|P\|\to 0} \sum_{k=1}^{n} \frac{1}{2}\big(f(\theta_k)\big)^2\,\Delta\theta_k = \int_{\alpha}^{\beta} \frac{1}{2}\big(f(\theta)\big)^2\,d\theta.$$

**Area of the Fan-Shaped Region Between the Origin and the Curve $r = f(\theta)$ when $\alpha \le \theta \le \beta$, $r \ge 0$, and $\beta - \alpha \le 2\pi$.**

$$A = \int_{\alpha}^{\beta} \frac{1}{2}r^2\,d\theta$$

This is the integral of the **area differential** (Figure 11.33)

$$dA = \frac{1}{2}r^2\,d\theta = \frac{1}{2}\big(f(\theta)\big)^2 d\theta.$$

**EXAMPLE 1** Find the area of the region in the $xy$-plane enclosed by the cardioid $r = 2(1 + \cos\theta)$.

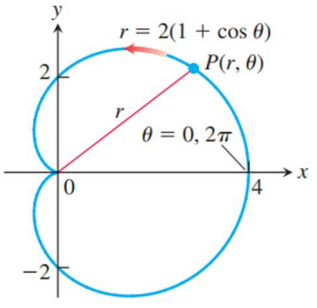



**FIGURE 11.34** The cardioid in Example 1.

**Area of the Region $0 \le r_1(\theta) \le r \le r_2(\theta)$, $\alpha \le \theta \le \beta$, and $\beta - \alpha \le 2\pi$.**

$$A = \int_\alpha^\beta \frac{1}{2} r_2^2 \, d\theta - \int_\alpha^\beta \frac{1}{2} r_1^2 \, d\theta = \int_\alpha^\beta \frac{1}{2} \left( r_2^2 - r_1^2 \right) d\theta \qquad (1)$$

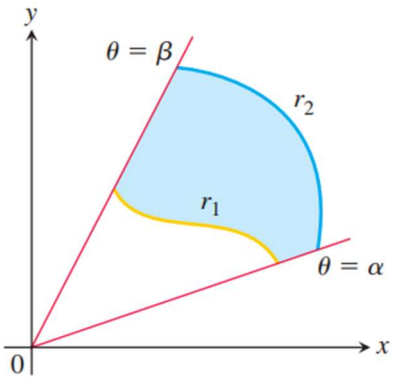



**FIGURE 11.35** The area of the shaded region is calculated by subtracting the area of the region between $r_1$ and the origin from the area of the region between $r_2$ and the origin.

**EXAMPLE 2** Find the area of the region that lies inside the circle $r = 1$ and outside the cardioid $r = 1 - \cos\theta$.

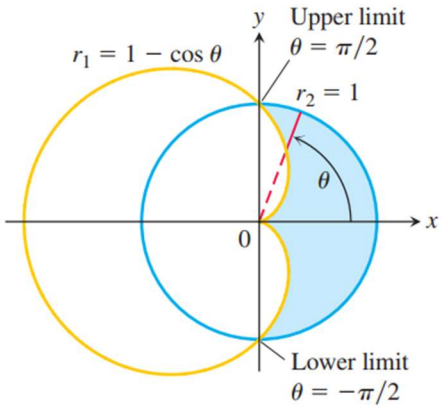



**FIGURE 11.36** The region and limits of integration in Example 2.

## Length of a Polar Curve

We can obtain a polar coordinate formula for the length of a curve $r = f(\theta)$, $\alpha \le \theta \le \beta$, by parametrizing the curve as

$$x = r\cos\theta = f(\theta)\cos\theta, \qquad y = r\sin\theta = f(\theta)\sin\theta, \qquad \alpha \le \theta \le \beta. \tag{2}$$

The parametric length formula, Equation (3) from Section 11.2, then gives the length as

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta.$$

This equation becomes

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

when Equations (2) are substituted for $x$ and $y$ (Exercise 29).

### Length of a Polar Curve

If $r = f(\theta)$ has a continuous first derivative for $\alpha \le \theta \le \beta$ and if the point $P(r, \theta)$ traces the curve $r = f(\theta)$ exactly once as $\theta$ runs from $\alpha$ to $\beta$, then the length of the curve is

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta. \tag{3}$$

**EXAMPLE 4** Find the length of the cardioid $r = 1 - \cos\theta$.

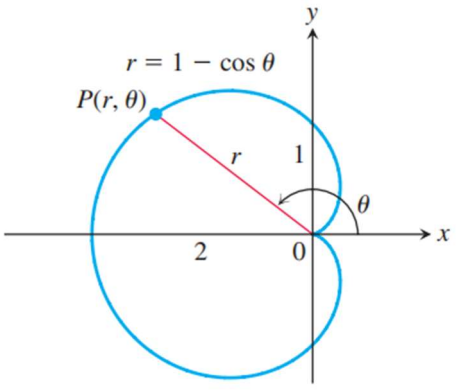


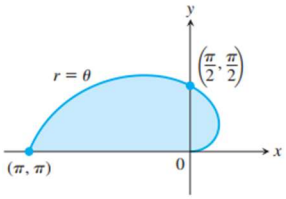

**FIGURE 11.38** Calculating the length of a cardioid (Example 4).

## EXERCISES 11.5

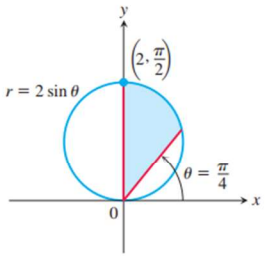
### Finding Polar Areas

Find the areas of the regions in Exercises 1–8.

**1.** Bounded by the spiral $r = \theta$ for $0 \le \theta \le \pi$




**2.** Bounded by the circle $r = 2 \sin \theta$ for $\pi/4 \le \theta \le \pi/2$




**3.** Inside the oval limaçon $r = 4 + 2 \cos \theta$

**4.** Inside the cardioid $r = a(1 + \cos \theta), \quad a > 0$

**5.** Inside one leaf of the four-leaved rose $r = \cos 2\theta$

**6.** Inside one leaf of the three-leaved rose $r = \cos 3\theta$



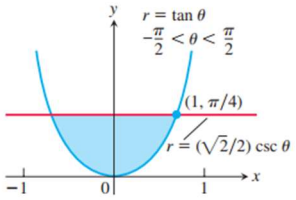

**15.** Inside the circle $r = -2 \cos \theta$ and outside the circle $r = 1$

**16.** Inside the circle $r = 6$ above the line $r = 3 \csc \theta$

**17.** Inside the circle $r = 4 \cos \theta$ and to the right of the vertical line $r = \sec \theta$

**18.** Inside the circle $r = 4 \sin \theta$ and below the horizontal line $r = 3 \csc \theta$

**19. a.** Find the area of the shaded region in the accompanying figure.




**b.** It looks as if the graph of $r = \tan \theta, -\pi/2 < \theta < \pi/2$, could be asymptotic to the lines $x = 1$ and $x = -1$. Is it? Give reasons for your answer.

**20.** The area of the region that lies inside the cardioid curve $r = \cos \theta + 1$ and outside the circle $r = \cos \theta$ is not

$$\frac{1}{2}\int_0^{2\pi} [(\cos \theta + 1)^2 - \cos^2 \theta]\, d\theta = \pi.$$

Why not? What *is* the area? Give reasons for your answers.

### Finding Lengths of Polar Curves

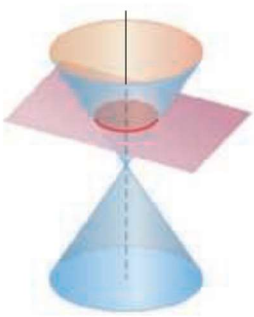
Find the lengths of the curves in Exercises 21–28.

**21.** The spiral $r = \theta^2, \quad 0 \le \theta \le \sqrt{5}$

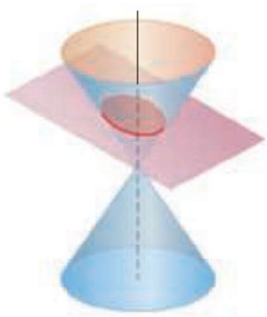
**22.** The spiral $r = e^\theta/\sqrt{2}, \quad 0 \le \theta \le \pi$

**23.** The cardioid $r = 1 + \cos \theta$

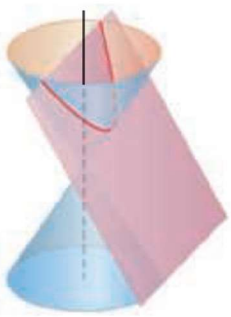
# 11.6 Conic Sections



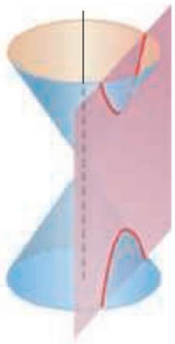
Circle: plane perpendicular to cone axis



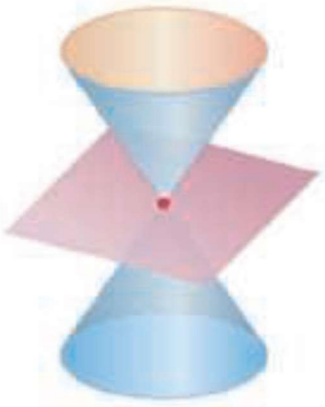
Ellipse: plane oblique to cone axis



Parabola: plane parallel to side of cone



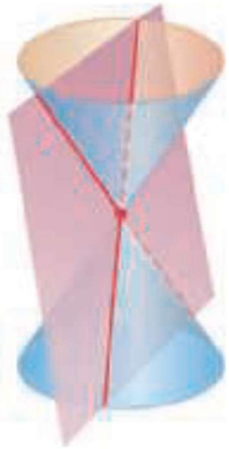
Hyperbola: plane parallel to cone axis



Point: plane through cone vertex only



Single line: plane tangent to cone



Pair of intersecting lines

## Parabolas

**DEFINITIONS** A set that consists of all the points in a plane equidistant from a given fixed point and a given fixed line in the plane is a **parabola**. The fixed point is the **focus** of the parabola. The fixed line is the **directrix**.

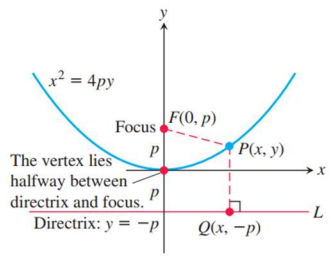



**FIGURE 11.40** The standard form of the parabola $x^2 = 4py$, $p > 0$.

$$PF = \sqrt{(x - 0)^2 + (y - p)^2} = \sqrt{x^2 + (y - p)^2}$$
$$PQ = \sqrt{(x - x)^2 + (y - (-p))^2} = \sqrt{(y + p)^2}.$$

When we equate these expressions, square, and simplify, we get

$$y = \frac{x^2}{4p} \quad \text{or} \quad x^2 = 4py. \qquad \text{Standard form} \qquad (1)$$

These equations reveal the parabola's symmetry about the $y$-axis. We call the $y$-axis the axis of the parabola (short for "axis of symmetry").

The point where a parabola crosses its axis is the **vertex**. The vertex of the parabola $x^2 = 4py$ lies at the origin (Figure 11.40). The positive number $p$ is the parabola's **focal length**.

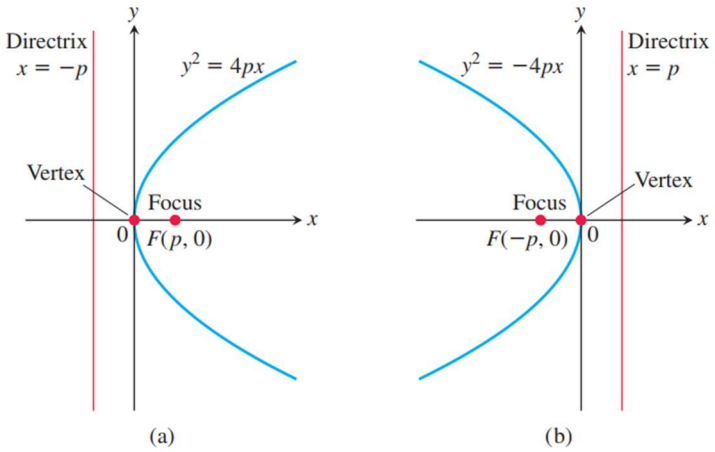



**FIGURE 11.41** (a) The parabola $y^2 = 4px$. (b) The parabola $y^2 = -4px$.

## Ellipses

**DEFINITIONS** An **ellipse** is the set of points in a plane whose distances from two fixed points in the plane have a constant sum. The two fixed points are the foci of the ellipse.

The line through the foci of an ellipse is the ellipse's **focal axis**. The point on the axis halfway be-tween the foci is the **center**. The points where the focal axis and ellipse cross are the ellipse's **vertices** (Figure 11.42).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

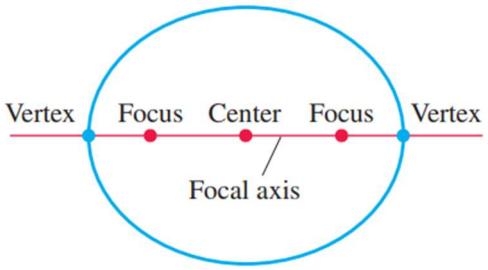



**FIGURE 11.42** Points on the focal axis of an ellipse.

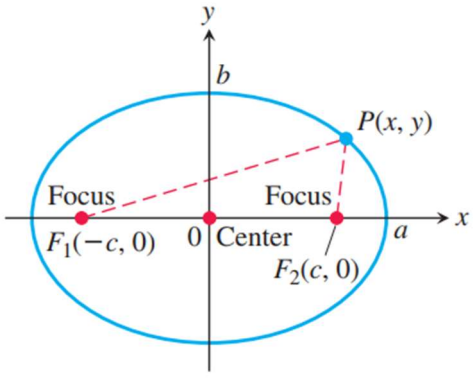



**FIGURE 11.43** The ellipse defined by the equation $PF_1 + PF_2 = 2a$ is the graph of the equation $(x^2/a^2) + (y^2/b^2) = 1$, where $b^2 = a^2 - c^2$.

## Standard-Form Equations for Ellipses Centered at the Origin

*Foci on the x-axis:* $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

*Foci on the y-axis:* $\dfrac{x^2}{b^2} + \dfrac{y^2}{a^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

In each case, $a$ is the semimajor axis and $b$ is the semiminor axis.

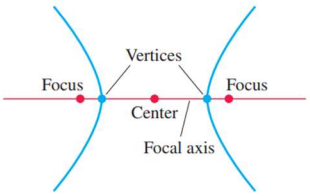



**FIGURE 11.45** Points on the focal axis of a hyperbola.

## Hyperbolas

**DEFINITIONS** A **hyperbola** is the set of points in a plane whose distances from two fixed points in the plane have a constant difference. The two fixed points are the foci of the hyperbola.

The line through the foci of a hyperbola is the **focal axis**. The point on the axis halfway between the foci is the hyperbola's **center**. The points where the focal axis and hyperbola cross are the **vertices** (Figure 11.45).

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

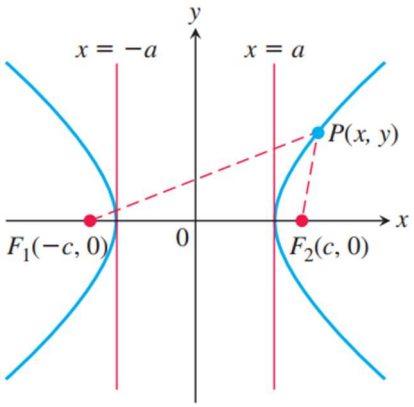



**FIGURE 11.46** Hyperbolas have two branches. For points on the right-hand branch of the hyperbola shown here, $PF_1 - PF_2 = 2a$. For points on the left-hand branch, $PF_2 - PF_1 = 2a$. We then let $b = \sqrt{c^2 - a^2}$.

**Standard-Form Equations for Hyperbolas Centered at the Origin**

*Foci on the x-axis:* $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$

Center-to-focus distance: $c = \sqrt{a^2 + b^2}$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

Asymptotes: $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 0$ or $y = \pm\dfrac{b}{a}x$

*Foci on the y-axis:* $\dfrac{y^2}{a^2} - \dfrac{x^2}{b^2} = 1$

Center-to-focus distance: $c = \sqrt{a^2 + b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

Asymptotes: $\dfrac{y^2}{a^2} - \dfrac{x^2}{b^2} = 0$ or $y = \pm\dfrac{a}{b}x$

Notice the difference in the asymptote equations ($b/a$ in the first, $a/b$ in the second).